

Multi-View Recognition of Driver Head Movements: A Measurement-First Study with Person-Held-Out Evaluation

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Abstract—Head movements are a primary observable of driver attention, and modern cabins increasingly carry several synchronized cameras. We study four-class recognition of head-movement direction (left/right/down/up) from three cabin views, reporting a systematic campaign of more than one hundred trained models on a 22-driver cohort. Our contributions are threefold. (1) *Method*: a parameter-efficient multi-view architecture that keeps a single Kinetics-pretrained video transformer frozen and learns only a small cross-view token-attention module (11M trainable parameters, 4% of one finetuned backbone). Under person-held-out evaluation it improves macro- F_1 by +6.3 points over an equally pretrained single-view model and +4.1 over trivial late fusion (both significant), halves the across-driver variance, and under a nested, selection-free protocol its micro-accuracy advantage over the single-view control grows to +5.1 points ($p=8.6 \times 10^{-12}$). (2) *Measurement*: we document how two common protocol defects—a video-level split that leaks driver identity into validation, and per-batch averaging of epoch metrics—jointly fabricated a coherent but false architecture narrative on this task; several “ablation-validated” conclusions, including our own earlier ones, reversed under a sound protocol. (3) *Negative results*: an analytic head-pose feature stream carries genuine, class-complementary signal in isolation, yet none of three injection routes (token-, query-, or logit-level) survived robustness checks, and six class-imbalance remedies failed to prevent rare-class collapse. We release the person-held-out protocol, paired statistical tooling, and all per-run predictions to support measurement-first research on small-cohort driver monitoring.

Index Terms—Driver monitoring systems, head movement recognition, multi-view fusion, video transformers, evaluation protocols, cross-subject generalization, negative results.

I. INTRODUCTION

DRIVER-monitoring systems (DMS) are becoming a regulatory requirement in new vehicles, and head behaviour is among their most informative observables: mirror checks, shoulder checks, and downward glances toward devices are all expressed as short, directed head movements. Compared with gaze estimation, head-movement classification is robust to eyewear and illumination and can be computed from standard cabin cameras. Modern cabins increasingly provide *several* such cameras, raising a natural question: how much does multi-view observation actually contribute to recognizing driver head behaviour, and through which fusion mechanism?

Answering this question rigorously is harder than it appears, for reasons that are structural to driver-monitoring research rather than incidental to our study. Cohorts are small—recruiting and recording drivers is expensive—so models see only tens of identities. Classes are heavily imbalanced, because natural driving produces many more downward and leftward glances than upward ones. And each driver contributes many highly correlated clips, so a split at the *clip* or *video* level leaks identity: a model can score well by recognizing *who* is moving rather than *how*. These conditions conspire to make small-cohort studies vulnerable to false conclusions that survive conventional ablation practice.

This paper is deliberately *measurement-first*. Our initial campaign of 40+ runs, conducted under a conventional protocol (random video-level split, default metric logging), produced what appeared to be a well-supported progression of architectural improvements, each confirmed by ablations. Auditing the protocol invalidated essentially all of it. We rebuilt the evaluation around person-held-out cross-validation with paired statistical tests, re-ran every claim, and report what survives—together with a controlled account of how the defective protocol manufactured the earlier narrative, and with the negative results of our attempts to exploit head-pose priors and to mitigate rare-class collapse.

Concretely, our contributions are:

- **MV-Min**, a minimal multi-view architecture: one frozen Kinetics-pretrained ViViT encoder shared across views, a per-view token compression to 17 tokens, and a small trainable transformer whose attention spans views. Under person-held-out evaluation it delivers significant macro- F_1 and robustness gains over equally pretrained single-view and late-fusion controls, and its advantage *grows* under a stricter nested protocol—the signature of a real effect rather than an artifact.
- **A sound evaluation protocol** for small-cohort driver monitoring—person-held-out k -fold with a nested selection-free variant, epoch-accumulated metrics, and paired McNemar/bootstrap tests—released with tooling, together with a quantified case study of how two silent protocol defects fabricated a false state-of-the-art narrative.
- **Documented negative results** on two fronts that the community is likely to attempt: injecting analytic head-pose priors (the signal is real, but no token-, query-,

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or logit-level route survived robustness checks) and six remedies for rare-class collapse (all failed). We argue that identity-invariant representation, not fusion architecture, is the binding constraint.

II. RELATED WORK

A. Driver monitoring and head behaviour

Vision-based driver monitoring spans distraction and secondary-task recognition [1], [2], [3], maneuver anticipation from cabin and road views [4], and head-pose estimation as a gaze proxy [5], [6]. Fine-grained benchmarks such as Drive&Act [1] and DMD [2] established multi-camera cabins as the standard setting and report cross-subject splits, but the interaction between tiny cohorts, class imbalance, and identity leakage has received little dedicated analysis. Our task—direction classification of spontaneous head movements—is narrower than activity recognition but isolates exactly the attention-relevant behaviour, and our cohort size (22 drivers) is typical of the regime where measurement pitfalls dominate.

B. Video backbones and parameter-efficient transfer

Video classification progressed from 3-D CNNs [7], [8], [9] to transformer architectures [10], [11], [12] pretrained on large corpora such as Kinetics [13]. For small downstream datasets, freezing the backbone and training light adapters is often preferable to full finetuning [14], [15]. Our results give this practice unusually sharp support: on 80 training videos, partial unfreezing contributed nothing measurable, while the pretrained representation contributed the single largest accuracy gain of the study.

C. Multi-view fusion

Multi-view action recognition classically fuses at the input, feature, or decision level [16]. Attention-based token fusion lets each view’s tokens attend to the others’, subsuming learned per-view weighting. Our contribution is not a new fusion operator but a controlled attribution: with backbone and pretraining held fixed, we measure what cross-view attention buys *beyond* late fusion on a small cohort—class balance and across-driver robustness rather than raw accuracy.

D. Evaluation pitfalls and negative results

Dataset bias and leakage are recognized threats to empirical validity [18], [19], and subject-dependent splits are known to inflate results in wearable activity recognition [20]. Statistical practice for classifier comparison is likewise well established [21], [22], [23] yet rarely applied in applied vision papers. We add a documented, quantified case in driver monitoring where two defects interacted to produce a self-consistent false narrative, and we normalize the reporting of negative results [24] for methods the community would plausibly retry.

III. TASK AND DATASET

A. Recording and annotation

Each of 22 drivers was recorded in four illumination conditions (day/night $\times$ high/low ambient light), yielding 88 sessions. Three synchronized cameras (front, left, right; 332 $\times$ 224 px) observe the cabin. Timeline annotations mark head-movement segments with one of eight direction labels, merged to four classes—*left*, *right*, *down*, *up*—with vertical components taking precedence (e.g. left-up $\rightarrow$ up). Unlabelled spans are excluded (no background class). Segment-level class frequencies are imbalanced: down 43%, left 28%, right 21%, up 9% (pooled over held-out folds).

Three annotators labelled every session. Frame-level inter-annotator agreement is 91.0% overall (399k co-labelled frames) and 94% for the *right* class. This matters for interpreting our results: the rare-class failures reported later cannot be attributed to label ambiguity.

A SAM-3D body estimator provides 70 keypoints (2-D and 3-D) per frame per view, of which eight are head-related (nose, eyes, ears, acromions, neck).

B. Why this task is statistically hostile

Three properties interact. (i) *Small cohort*: 22 identities upper-bound what any split can test. (ii) *Correlated clips*: each session yields tens of segments of the same person in the same seat, so clip- or video-level splits measure identity recall. (iii) *Imbalance*: the majority class alone achieves 42.7% accuracy. A single 8-video validation split—the protocol of our own first campaign—carries a binomial 95% CI of ± 6.4 points and, as we later measured, up to 15.6 points of across-fold spread. Every architectural effect we chased in four iterations was smaller than that noise.

IV. EVALUATION PROTOCOL

We treat the protocol as a first-class contribution and specify it fully.

Person-held-out 5-fold CV. The 22 drivers are partitioned into five folds (4–5 drivers each, seeded). For fold k , all sessions of the fold’s drivers form the evaluation set; the model trains on the remaining drivers. Pooling folds scores every driver exactly once with a model that never saw them ($n=3954$ segments).

Nested selection-free variant. Checkpoint selection itself can leak: choosing the best epoch on the same people one reports on inflated our arms by +0.9 to +2.0 points (measured). In the nested protocol, three drivers are further carved out of each training set as an inner validation set; checkpoints are selected on the inner set and reported on the held-out fold. Train, inner-validation, and test sets are person-disjoint by construction.

Metrics. micro-accuracy (segment accuracy), macro-accuracy, and macro- F_1 , all accumulated over the full epoch and reduced once. We report macro- F_1 alongside micro-accuracy because the majority class alone reaches 42.7% micro-accuracy; balance is the operationally relevant axis for a monitoring system.

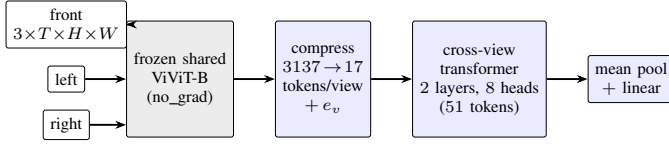


Fig. 1. MV-Min: one frozen Kinetics-pretrained ViViT encoder shared by all views; only the shaded-blue modules train (11M parameters).

Statistics. All model comparisons are *paired* on identical segment sets (verified fold-by-fold before testing): McNemar’s test with continuity correction [21], [22] for micro-accuracy, and a paired bootstrap (1,500–2,000 resamples, 95% CIs) [23] for all metrics. The pooled protocol’s binomial CI is ± 1.6 points; we treat sub-CI differences as noise regardless of nominal p -values.

Training. All arms: 50 epochs, no early stopping, Adam (lr 10^{-4} , cosine decay), effective batch 16, bf16 mixed precision, class-frequency weights in the cross-entropy loss, one H100 per run. ViViT arms sample 32 frames at 224^2 ; 3-D CNN arms 8 frames.

V. METHOD: MV-MIN

The architecture that survives evaluation is deliberately minimal (Fig. 1). Let $V = \{\text{front, left, right}\}$ and let $x_v \in \mathbb{R}^{3 \times T \times H \times W}$ be view v ’s clip.

Frozen shared encoder. A single ViViT-B/16 $\times$ 2 encoder [11], pretrained on Kinetics-400 [13], processes each view independently with shared (frozen) weights, producing token matrices $Z_v \in \mathbb{R}^{(1+N_t N_s) \times D}$ with $N_t=16$ temporal tubelets, $N_s=196$ spatial patches, $D=768$. The encoder runs under `no_grad`; no backbone parameter is updated.

Token compression. Attention over 3×3137 tokens is wasteful and, on 80 training videos, an overfitting surface. Per view we keep $1+N_t=17$ tokens: the CLS token and one spatially averaged token per tubelet,

$$z_v^{(t)} = \frac{1}{N_s} \sum_{s=1}^{N_s} Z_v^{(t,s)}, \quad t = 1, \dots, N_t. \quad (1)$$

Cross-view fusion. Each view’s tokens receive a learned additive view embedding $e_v \in \mathbb{R}^D$; the $3 \times 17=51$ tokens are concatenated into one sequence and processed by a 2-layer, 8-head transformer encoder [17] whose self-attention spans all views. Mean pooling and a linear head produce the 4-class logits.

Capacity. Trainable parameters: 11M (fusion module and head), i.e. 4% of one finetuned ViViT-B; multi-view inference reuses the single frozen encoder three times.

Three design decisions deserve emphasis because each contradicts what our own leaky-protocol ablations had “established”: the backbone is fully frozen (partial unfreezing measured n.s. under the sound protocol), no keypoint stream is included (Sec. VIII), and aggregation is plain mean pooling (learned gating hurt in every protocol we tried).

VI. MAIN RESULTS

Tables I and II support three statements.

TABLE I
PERSON-HELD-OUT 5-FOLD CV (POOLED $n=3954$; SEED 42; %).
“SPREAD” IS THE ACROSS-FOLD STD. OF MICRO-ACCURACY.

Model	micro	mac. acc	macro-F ₁	spread
Majority class	42.7	25.0	15.0	–
3D-CNN, single view	47.0	31.1	26.8	.156
TS-CVA mid fusion (3D-CNN)	45.4	31.9	28.1	–
ViViT, single view	52.5	35.5	29.8	.134
ViViT, 3-view late fusion	52.6	35.7	31.9	.114
MV-Min (ours)	51.5	38.0	36.5	.090

TABLE II
PAIRED COMPARISONS OF THE FULL MULTI-VIEW ARM ON IDENTICAL SEGMENTS ($n=3954$; POINTS; 95% BOOTSTRAP CIs; MCNEMAR p FOR MICRO-ACCURACY).

Comparison	Δ micro	Δ macro-F ₁	p
vs. 3D-CNN single	+5.1 [+3.6, +6.6]	+9.2	1.9×10^{-11}
vs. ViViT single	−0.4 [−1.7, +0.9]	+6.3 [+4.8, +7.7]	0.55
vs. ViViT late	−0.4 [−1.8, +0.9]	+4.1 [+2.4, +5.8]	0.55

(1) Pretraining dominates micro-accuracy. Replacing the 3-D CNN with a Kinetics-pretrained ViViT adds +5.5 micro-accuracy points—the largest single effect measured in the entire campaign. No fusion mechanism adds micro-accuracy on top of it under the standard protocol. Attribution required the single-view ViViT control; without it, the +5.1 of the full multi-view arm over the 3-D CNN baseline would have been misread as a multi-view contribution.

(2) Cross-view attention buys balance and robustness. Against the equally pretrained single-view and trivial late-fusion controls, cross-view token attention is micro-accuracy-neutral but gains +6.3 and +4.1 macro-F₁ respectively (both significant) and reduces the across-driver spread from .134 to .090. For a monitoring system, missing rare-but-critical movements (*up*: mirror checks) and failing on unfamiliar drivers are the relevant failure modes; this is the axis on which multi-view observation demonstrably helps.

(3) A previously winning fusion reverses. TS-CVA mid fusion, the best fusion of our leaky-protocol campaign (48.9 vs. 37.2 single view), scores *below* its own single-view baseline under person hold-out (45.4 vs. 47.0). Fusion conclusions drawn under identity leakage did not merely shrink—they changed sign.

A. Nested, selection-free evaluation

Under the nested protocol (Table III) every number drops, but the ordering sharpens in our favour: MV-Min is the best arm, and the paired advantage of cross-view attention over the single-view control *grows* to +5.1 micro-accuracy ($p=8.6 \times 10^{-12}$; macro-F₁ +2.9 [+1.4, +4.4]). A result that strengthens as selection freedom is removed is the signature of a real effect; the two arms that weaken (partial unfreezing, pose tokens) are excluded from the method accordingly.

B. Seed variance

Replicating both headline arms with seeds 42/43/44: single-view ViViT 52.5/53.0/51.9 micro-accuracy; MV full

TABLE III
NESTED PROTOCOL: CHECKPOINTS SELECTED ON INNER
(PERSON-DISJOINT) VALIDATION DRIVERS; REPORTED ON HELD-OUT
DRIVERS ($n=3954$; %).

Arm	micro	macro-F ₁
ViViT, single view	44.7	28.8
MV full (unfreeze+kpt)	49.8	31.8
MV-Min	50.6	—
+ pose tokens (Sec. VIII)	48.1	—

TABLE IV
COMPONENT ATTRIBUTION UNDER PERSON-HELD-OUT CV. PAIRED
EFFECTS ARE VS. MV-MIN UNLESS NOTED; “N.S.” AT $\alpha=.05$.

Arm	micro	macro-F ₁	paired effect
MV-Min (reference)	51.5	36.5	—
+ partial unfreeze (4 layers)	52.4	36.2	n.s./n.s.
+ kpt-query pooling	52.1	36.0	+1.3*/n.s.
+ unfreeze depth 6	50.5	35.5	n.s.
+ mirror augmentation	53.0	35.3	n.s.
raw-kpt token stream	51.2	36.4	n.s.

*McNemar $p=0.004$; reverses under the nested protocol.

52.1/52.6/51.9. Run-to-run variation (± 0.6) sits well inside the protocol CI, and the macro-F₁ advantage of cross-view fusion persists in all three seeds (mean +5.0).

VII. ATTRIBUTION ABLATION

No addition to the minimal form produces a paired improvement that is robust across both metrics and both protocols (Table IV). Under the leaky protocol, several of these same components had appeared decisively beneficial (“query pooling +3.5”, “optimal unfreeze depth 6”, “partial unfreezing +2.9”). The minimal form is therefore the published method: everything else failed attribution.

VIII. HEAD-POSE PRIORS: REAL SIGNAL, NO ROBUST INJECTION ROUTE

Head movements should be readable from head pose, and the scenario provides per-frame 3-D keypoints. This section documents why naive keypoint fusion failed, evidence that the signal itself is real, and the failure of three injection routes—results we report so that others need not rediscover them.

A. Why raw keypoint streams fail

The raw 70×3 tensor is dominated by hands: 42 of 70 rows (60% of input dimensions) are hand points whose frame-to-frame motion is $4 \times$ that of the head, leaving the head with $\sim 3\%$ of the input’s motion energy. Coordinates are unnormalized camera coordinates, so the largest variance directions encode *who and where the driver is*—precisely the identity signal a person-held-out model must ignore. And the earlier design compressed the trajectory into a single token competing with 51 video tokens. The head signal never actually reached the model; the corresponding arms measured n.s. accordingly.

B. Analytic head-pose features

We compute what the network was supposed to learn. With $p_t^{(i)}$ the 3-D position of keypoint i at frame t , define the ear midpoint $o_t = \frac{1}{2}(p_t^{\text{le}} + p_t^{\text{re}})$, ear axis $\mathbf{e}_t = p_t^{\text{le}} - p_t^{\text{re}}$, and facing direction $\mathbf{f}_t$ as the nose direction orthogonalized against $\mathbf{e}_t$ (bars denote unit normalization). Head angles follow analytically:

$$\begin{aligned} \text{yaw}_t &= \text{atan2}(f_{t,x}, |f_{t,z}|), & \text{pitch}_t &= \text{atan2}(f_{t,y}, \|f_{t,xz}\|), \\ \text{roll}_t &= \text{atan2}(e_{t,y}, \|e_{t,xz}\|). \end{aligned} \quad (2)$$

The per-frame feature (33-D) concatenates the three angles, their first differences (the label is a *movement*—a derivative), the head-centroid velocity, and the eight head points expressed in the local frame $(\mathbf{e}_t, \mathbf{f}_t, \mathbf{e}_t \times \mathbf{f}_t)$; angles and local coordinates are centred by their per-segment temporal mean, removing each driver’s resting posture (identity) while preserving the movement.

C. The signal is real

A 64-unit MLP on these features alone—no video—reaches 27.2 macro-F₁ on unseen drivers with *non-zero F₁ on every class*: right 0.24 and up 0.12, where every video model scores ≈ 0.03 and 0.00 (Table V). Within a clip, yaw spans up to 69° ; the prior is strong and complementary to the video models’ failure modes.

D. Three injection routes, none robust

Token-level. Per-frame pose tokens (one per frame per view, projected to D and tagged with a stream embedding) initially produced the best micro-accuracy of the campaign (53.3, +1.7 over MV-Min, $p=8.9 \times 10^{-5}$). Three pre-registered robustness checks demoted it: seed sensitivity (+1.7/ +3.3/ +0.4; significant in 2 of 3); *reversal* under the nested protocol (-2.5 , $p=3.2 \times 10^{-7}$), showing part of the gain came from checkpoint selection on scored drivers; and a representation control in which raw flattened keypoints in the same per-frame token format captured most of the effect (+1.1, $p=0.021$), leaving the analytic encoding itself at +0.6 (n.s.). **Query-level.** Using the pose trajectory as the cross-attention query for pooling: +1.3 micro-accuracy ($p=0.004$) but macro-F₁-neutral and not robust under nesting (Table IV). **Logit-level.** A pre-registered 50/50 probability average of the pose-only probe and the video model degraded micro-accuracy by 9.1 points ($p=5.4 \times 10^{-30}$): the probe is 22 points weaker overall, and the equal-weight blend loses more on frequent classes than it recovers on rare ones. An α -sweep shows a mild trade-off region near $\alpha=0.8$ (up-F₁ 0.02 $\rightarrow$ 0.06 at $\approx$ flat micro-accuracy), which we report as sensitivity only: selecting α on test folds would reintroduce selection bias.

The open problem is precise: an injection mechanism that preserves the probe’s rare-class competence without surrendering the video model’s frequent-class strength. Learning the blend on nested inner-validation drivers is the legitimate next step.

TABLE V
PER-CLASS F_1 UNDER PERSON-HELD-OUT CV. EVERY VIDEO MODEL COLLAPSES ONTO FREQUENT CLASSES; ONLY THE POSE-ONLY PROBE PREDICTS ALL FOUR.

Model	left	right	down	up
MV-Min	.538	.173	.673	.078
+ pose tokens	.532	.179	.695	.015
Pose-only probe (33-D)	.327	.237	.409	.117

TABLE VI
REMEDIES FOR CLASS COLLAPSE UNDER THE SOUND PROTOCOL. NONE SUCCEEDED.

Remedy	micro	outcome
Class-weighted CE (default)	51.5	collapse persists
Focal loss [25]	46.1 [‡]	harmful (−5.4)
Balanced softmax [26]	26.4	catastrophic
Clip-mean subtraction	43.6	harmful
Pose-token fusion	53.3 [†]	up- F_1 drops
Probe-model logit fusion (50/50)	44.1	harmful

[†]not robust (Sec. VIII); [‡]leaky-protocol measurement, direction consistent.

IX. RARE-CLASS COLLAPSE AND FAILED REMEDIES

Two failure modes define the actual difficulty of this task.

Collapse. *Right* and *up* are essentially never predicted by any video model (Table V)—the best model answers 1 of 59 *right* segments correctly in the worst fold—despite 94% annotator agreement on *right*. Six remedies spanning loss design (focal [25], balanced softmax [26], class weighting [27]), input design (clip-mean subtraction), and fusion design (pose tokens, logit fusion) all failed (Table VI).

Driver memorisation. Under person hold-out, the best checkpoint lands at epoch 0–4 in *every* arm: models begin memorising training drivers almost immediately, consistent with the large across-driver spread (fold-level micro-accuracy ranges from 21% to 63% for the 3-D CNN baseline). Clip-mean subtraction, our direct attack on static appearance, removed useful signal along with identity. We conclude that identity-invariant representation—not fusion architecture—is the binding constraint for small-cohort driver monitoring, and nominate it as the field’s priority.

X. HOW A DEFECTIVE PROTOCOL FABRICATED A NARRATIVE

We document this because both defects are common, silent, and mutually reinforcing.

Defect 1: identity leakage. Our first campaign split at the video level; all seven validation drivers also appeared in training. Scores measured recall of known drivers: the same 3-D CNN drops from 53.6% (leaky) to 47.0% (person-held-out), and fold-level results vary by 15.6 points depending on which drivers are held out.

Defect 2: per-batch metric averaging. Epoch metrics were computed per batch (size 2) and averaged—inflating macro- F_1 by up to 17 points (0.52 logged vs. 0.35 true)—and, worse, *inconsistently*: one trainer family accumulated correctly, so cross-architecture comparisons mixed two incompatible scales.

Together these produced a coherent and false progression (54.0 $\rightarrow$ 56.9 $\rightarrow$ 58.3 “state of the art”), each step confirmed by ablations run under the same defects; an intermediate “architectures are indistinguishable” reading, drawn from the same leaky split, was equally wrong. Under the sound protocol the query-pooling effect (+3.5) vanished, the unfreeze-depth sweep became n.s., and the TS-CVA fusion champion fell below its own baseline. Nothing about these defects is exotic—one is a dataloader default, the other a metric-API misuse—which is precisely why we report the case quantitatively and release the protocol as the antidote.

XI. DISCUSSION AND LIMITATIONS

What multi-view is for. On this task, multi-view observation does not raise headline accuracy over a strong pretrained single-view model; it raises class balance and cross-driver stability. Deployments should choose multi-view for robustness, not for accuracy benchmarks.

Scope. Our cohort (22 drivers, one vehicle type, four illumination conditions) matches the regime we analyse but limits external validity; cross-dataset evaluation (e.g. Drive&Act [1], DMD [2]) is future work. Segment-level classification with oracle temporal boundaries sidesteps detection; an online system must also localize movements. The keypoint study depends on one 3-D body estimator; better head-dedicated pose estimators [5], [6] may change the injection results. Finally, our negative results are task-scoped: we show these remedies fail *here*, not everywhere.

Recommendations for small-cohort DMS research. (1) Split by person; report pooled person-held-out results with per-fold spread. (2) Select checkpoints on person-disjoint inner validation. (3) Accumulate metrics over the epoch; verify metric parity across baselines. (4) Pair every claim with McNemar/bootstrap statistics on identical samples. (5) Report macro- F_1 beside micro-accuracy whenever a majority class exceeds a third of the data. (6) Publish per-run predictions so later work can re-test claims.

XII. CONCLUSION

Under person-held-out, statistically tested evaluation of more than twenty experimental arms, the defensible findings on multi-view driver head-movement recognition are: (1) Kinetics-pretrained video transformers dominate accuracy (+5.5 micro-accuracy over a 3-D CNN); (2) a frozen-backbone cross-view token-attention module with 4% of the trainable parameters of one finetuned backbone adds class balance (+6.3 macro- F_1), halves across-driver spread, and, under the strictest selection-free protocol, adds +5.1 micro-accuracy ($p=8.6 \times 10^{-12}$) over its single-view control; (3) head-pose priors carry real, class-complementary signal but no injection route yet survives robustness checks; (4) rare-class collapse and driver memorisation, not fusion architecture, are the binding constraints—six remedies failed; (5) two ordinary measurement defects sufficed to fabricate, and ablations to “confirm”, an entire false research narrative. We release the protocol, statistical tooling, and per-run predictions.

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To be completed.

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